

A. Introduction

1. **Title:** Cyber Security — Information Protection
2. **Number:** RRC-CIP-011-1
3. **Purpose:** To prevent unauthorized access to BES Cyber System Information (BCSI) by specifying information protection requirements in support of protecting BES Cyber Systems (BCS) against compromise that could result in instability, uncontrolled separations, or Cascading within the Railbelt Region BES.
4. **Applicability:**
 - 4.1. **Functional Entities:** For the purpose of the requirements contained herein, the following list of functional entities will be collectively referred to as “Responsible Entities.”
 - 4.1.1. **Balancing Authority**
 - 4.1.2. **Generator Operator**
 - 4.1.3. **Generator Owner**
 - 4.1.4. **Transmission Operator**
 - 4.1.5. **Transmission Owner**
 - 4.1.6. **Distribution Provider:** that owns one or more of the following Facilities, systems, and equipment for the protection or restoration of the BES:
 - 4.1.6.1. Each underfrequency load shedding (UFLS) or undervoltage loadshedding (UVLS) system that:
 - 4.1.6.1.1. is part of a Load shedding program that is subject to one or more requirements in an Railbelt Reliability Council Standard; and
 - 4.1.6.1.2. performs automatic Load shedding under a common control system owned by the Responsible Entity, without human operator initiation, that the loss, compromise, or misuse could have an adverse impact on the reliable operation of the BES.
 - 4.1.6.2. Each Remedial Action Scheme (RAS) where the RAS is subject to one or more requirements in an Railbelt Reliability Council Standard.
 - 4.1.6.3. Each Protection System (excluding UFLS and UVLS) that applies to Transmission where the Protection System is subject to one or more requirements in an Railbelt Reliability Council Standard.
 - 4.1.6.4. Each Cranking Path and group of Elements meeting the initial switching requirements from a Blackstart Resource up to and including the first interconnection point of the starting station service of the next generation unit(s) to be started.
 - 4.2. **Facilities:** For the purpose of the requirements contained herein, the following Facilities, systems, and equipment owned by each Responsible Entity in 4.1 above are those to which these requirements are applicable. For requirements in this standard where a specific type of Facilities, system, or equipment or subset of Facilities, systems, and equipment are

applicable, these are specified explicitly.

4.2.1. Distribution Provider: One or more of the following Facilities, systems and equipment owned by the Distribution Provider for the protection or restoration of the BES:

4.2.1.1. Each UFLS or UVLS System that:

4.2.1.1.1. is part of a Load shedding program that is subject to one or more requirements in an Railbelt Reliability Council Standard; and

4.2.1.1.2. performs automatic Load shedding under a common control system owned by the Responsible Entity, without human operator initiation, that the loss, compromise, or misuse of could have adverse impact on the reliable operation of the BES.

4.2.1.2. Each RAS where the RAS is subject to one or more requirements in an Railbelt Reliability Council Standard.

4.2.1.3. Each Protection System (excluding UFLS and UVLS) that applies to Transmission where the Protection System is subject to one or more requirements in an Railbelt Reliability Council Standard.

4.2.1.4. Each Cranking Path and group of Elements meeting the initial switching requirements from a Blackstart Resource up to and including the first interconnection point of the starting station service of the next generation unit(s) to be started.

4.2.2. Responsible Entities listed in 4.1 other than Distribution Providers:

All BES Facilities.

4.2.3. Exemptions: The following are exempt from Standard **RRC-CIP-011-1**:

4.2.3.1. Cyber Systems at Facilities regulated by the Canadian Nuclear Safety Commission.

4.2.3.2. Cyber Assets associated with communication networks and data communication links between discrete Electronic Security Perimeters (ESP).

4.2.3.3. Cyber Systems, associated with communication networks and data communication links, between Cyber Systems, providing confidentiality and integrity of an ESP that extends to one or more geographic locations.

4.2.3.4. The systems, structures, and components that are regulated by the Nuclear Regulatory Commission under a cyber security plan pursuant to 10 C.F.R. Section 73.54.

4.2.3.5. For Distribution Providers, the systems and equipment that are not included in section 4.2.1 above.

4.2.3.6. Responsible Entities that identify that they have no BES Cyber Systems categorized as high impact or medium impact according

to the RRC-CIP-002-1 identification and categorization processes.

- 4.3. “Applicable Systems”:** Each table has an “Applicable Systems” column to define the scope of the systems to which a specific requirement part applies.
- 5. Effective Dates:** Shall become effective on the first day of the first calendar quarter, that is twenty-four (24) months following adoption by the Approving Authority.

B. Requirements and Measures

- R1.** Each Responsible Entity shall implement one or more documented information protection program(s) that collectively includes each of the applicable requirement parts in *RRC-CIP-011-1 Table R1 – Information Protection*. [Violation Risk Factor: Medium]
- M1.** Evidence for the information protection program must include the applicable requirement parts in *RRC-CIP-011-1 Table R1 – Information Protection* and additional evidence to demonstrate implementation as described in the Measures column of the table.

RRC-CIP-011-1 Table R1 – Information Protection Program			
Part	Applicable Systems	Requirements	Measures
1.1	High Impact BCS and their associated: 1. Electronic Access Control and Monitoring Systems (EACMS); and 2. Physical Access Control Systems (PACS) Medium Impact BCS and their associated: 1. EACMS; and 2. PACS Shared Cyber Infrastructure (SCI) supporting an Applicable System in this Part	Method(s) to identify BCSl.	Examples of acceptable evidence include, but are not limited to: • Documented method to identify BCSl from entity's information protection program; or • Indications on information (e.g., labels or classification) that identify BCSl as designated in the entity's information protection program; or • Training materials that provide personnel with sufficient knowledge to identify BCSl; or • Storage locations identified for housing BCSl in the entity's information protection program.

RRC-CIP-011-1 Table R1 – Information Protection Program			
Part	Applicable Systems	Requirements	Measures
1.2	High Impact BCS and their associated: 1. EACMS; and 2. PACS Medium Impact BCS and their associated: 1. EACMS; and 2. PACS SCI supporting an Applicable System in this Part	Method(s) to protect and securely handle BCSI to mitigate risks of compromising confidentiality.	Examples of acceptable evidence for on-premise include, but are not limited to: • Procedures for protecting and securely handling, which include topics such as storage, security during transit, and use of BCSI; or • Records indicating that BCSI is handled in a manner consistent with the entity's documented procedure(s). Examples of evidence for off-premise BCSI may include, but are not limited to, the following: • Implementation of electronic technical method(s) to protect electronic BCSI (e.g., data masking, encryption, hashing, tokenization, cipher, electronic key management); or • Implementation of physical technical method(s) to protect physical BCSI (e.g., physical lock and key management, physical badge management, biometrics, alarm system); or

RRC-CIP-011-1 Table R1 – Information Protection Program			
Part	Applicable Systems	Requirements	Measures
			• Implementation of administrative method(s) to protect BCSI (e.g., vendor service risk assessments, business agreements).

- R2.** Each Responsible Entity shall implement one or more documented process(es) that collectively include the applicable requirement parts in *RRC-CIP-011-1 Table R2 – Reuse and Disposal*. [Violation Risk Factor: Lower]
- M2.** Evidence must include each of the applicable documented processes that collectively include each of the applicable requirement parts in *RRC-CIP-011-1 Table R2 – Reuse and Disposal* and additional evidence to demonstrate implementation as described in the Measures column of the table.

RRC-CIP-011-1 Table R2 – Reuse and Disposal			
Part	Applicable Systems	Requirements	Measures
2.1	High Impact BCS and their associated: 1. EACMS; 2. PACS; and 3. Protected Cyber Asset (PCA) Medium Impact BCS and their associated: 1. EACMS; 2. PACS; and 3. PCA SCI supporting an Applicable System in this Part	Methods to prevent the unauthorized retrieval of BCSI from Applicable Systems containing BCSI, prior to their disposal or reuse (except for reuse within other systems identified in the Applicable Systems column).	Examples of acceptable evidence include, but are not limited to: • Records tracking sanitization actions taken to prevent unauthorized retrieval of BCSI such as clearing, purging, or destroying; or • Records tracking actions such as encrypting, retaining in the Physical Security Perimeter (PSP) or other methods used to prevent unauthorized retrieval of BCSI.

C. Compliance

1 Compliance Monitoring Process

1.1. Compliance Enforcement Authority:

The RRC is the entity with the authority to monitor and enforce compliance with mandatory and enforceable Railbelt Reliability Council Standards.

1.2. Evidence Retention:

The following evidence retention periods identify the period of time an entity is required to retain specific evidence to demonstrate compliance. For instances where the evidence retention period specified below is shorter than the time since the last audit, the RRC may ask an entity to provide other evidence to show that it was compliant for the full time period since the last audit.

The Responsible Entity shall keep data or evidence to show compliance as identified below unless directed by its RRC to retain specific evidence for a longer period of time as part of an investigation:

- Each Responsible Entity shall retain evidence of each requirement in this standard for three calendar years.
- If a Responsible Entity is found non-compliant, it shall keep information related to the non-compliance until mitigation is complete and approved or for the time specified above, whichever is longer.
- The RRC shall keep the last audit records and all requested and submitted subsequent audit records.

1.3. Compliance Monitoring and Enforcement Program:

Compliance Monitoring and Enforcement Program refers to the identification of the processes that will be used to evaluate data or information for the purpose of assessing performance or outcomes with the associated Railbelt Reliability Council Standards.

Table of Compliance Elements

R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-011-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
R1	Medium	N/A	N/A	The Responsible Entity did not implement one or more BCSI protection program(s). (R1) OR The Responsible Entity did not implement at least one method to identify BCSI. (1.1) OR The Responsible Entity did not implement at least one method to protect and securely handle BCSI. (1.2)	The Responsible Entity neither documented nor implemented one or more BCSI protection program(s). (R1)
R2	Lower	N/A	The Responsible Entity did not include processes for reuse to prevent the unauthorized retrieval of BCSI from an Applicable System. (2.1)	The Responsible Entity did not include disposal processes to prevent the unauthorized retrieval of BCSI from an Applicable System. (2.1)	The Responsible Entity neither documented nor implemented any processes for applicable requirement parts in RRC-CIP-011-1 Table R2 – Reuse and Disposal. (R2)

D. Regional Variances

None.

E. Interpretations

None.

F. Associated Documents

None.

Version History

Version	Date	Action	Change Tracking

Balancing Authority	The responsible entity that integrates resource plans ahead of time, maintains Demand and resource balance within a Balancing Authority Area, and supports Interconnection frequency in real time.
BES Cyber Asset (BCA)	A Cyber Asset that if rendered unavailable, degraded, or misused would, within 15 minutes of its required operation, misoperation, or non operation, adversely impact one or more Facilities, systems, or equipment, which, if destroyed, degraded, or otherwise rendered unavailable when needed, would affect the reliable operation of the Bulk Electric System. Redundancy of affected Facilities, systems, and equipment shall not be considered when determining adverse impact. Each BES Cyber Asset is included in one or more BES Cyber Systems.
BES Cyber System (BCS)	One or more BES Cyber Assets logically grouped by a responsible entity to perform one or more reliability tasks for a functional entity.
BES Cyber System Information	Information about the BES Cyber System that could be used to gain unauthorized access or pose a security threat to the BES Cyber System. BES Cyber System Information does not include individual pieces of information that by themselves do not pose a threat or could not be used to allow unauthorized access to BES Cyber Systems, such as, but not limited to, device names, individual IP addresses without context, ESP names, or policy statements. Examples of BES Cyber System Information may include, but are not limited to, security procedures or security information about BES Cyber Systems, Physical Access Control Systems, and Electronic Access Control or Monitoring Systems that is not publicly available and could be used to allow unauthorized access or unauthorized distribution; collections of network addresses; and network topology of the BES Cyber System.
Blackstart Resource	A generating unit(s) and its associated set of equipment which has the ability to be started without support from the System or is designed to remain energized without connection to the remainder of the System, with the ability to energize a bus, meeting the Transmission Operator's restoration plan needs for Real and Reactive Power capability, frequency and voltage control, and that has been included in the Transmission Operator's restoration plan.
Bulk Electric System	Unless modified by the lists shown below, all Transmission Elements operated at 100 kV or higher and Real Power and Reactive Power resources connected at 100 kV or higher. This does not include facilities used in the local distribution of electric energy. Inclusions: • I1 - Transformers with the primary terminal and at least one secondary terminal operated at 100 kV or higher unless excluded by application of

Exclusion E1 or E3. • I2 – Generating resource(s) including the generator terminals through the high-side of the step-up transformer(s) connected at a voltage of 100 kV or above with: a) Gross individual nameplate rating greater than 20 MVA. Or, b) Gross plant/facility aggregate nameplate rating greater than 75 MVA. • I3 – Blackstart Resources identified in the Transmission Operator’s restoration plan. • I4 – Dispersed power producing resources that aggregate to a total capacity greater than 75 MVA (gross nameplate rating), and that are connected through a system designed primarily for delivering such capacity to a common point of connection at a voltage of 100 kV or above. Thus, the facilities designated as BES are: a) The individual resources, and The system designed primarily for delivering capacity from the point where those resources aggregate to greater than 75 MVA to a common point of connection at a voltage of 100 kV or above. • I5 –Static or dynamic devices (excluding generators) dedicated to supplying or absorbing Reactive Power that are connected at 100 kV or higher, or through a dedicated transformer with a high-side voltage of 100 kV or higher, or through a transformer that is designated in Inclusion I1 unless excluded by application of Exclusion E4 Exclusions: • E1 – Radial systems: A group of contiguous transmission Elements that emanates from a single point of connection of 100 kV or higher and: a) Only serves Load. Or, b) Only includes generation resources, not identified in Inclusions I2, I3, or I4, with an aggregate capacity less than or equal to 75 MVA (gross nameplate rating). Or, c) Where the radial system serves Load and includes generation resources, not identified in Inclusions I2, I3 or I4, with an aggregate capacity of non-retail generation less than or equal to 75 MVA (gross nameplate rating). Note 1 – A normally open switching device between radial systems, as depicted on prints or one-line diagrams for example, does not affect this exclusion. Note 2 – The presence of a contiguous loop, operated at a voltage level of 50 kV or less, between configurations being considered as radial systems, does not affect this exclusion. • E2 – A generating unit or multiple generating units on the customer’s side of the retail meter that serve all or part of the retail Load with electric energy if: (i) the net capacity provided to the BES does not exceed 75 MVA, and (ii) standby, back-up, and maintenance power services are provided to the generating unit or multiple generating units or to the retail Load by a Balancing Authority, or provided pursuant to a binding obligation with a
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	Generator Owner or Generator Operator, or under terms approved by the applicable regulatory authority. • E3 - Local networks (LN): A group of contiguous transmission Elements operated at less than 300 kV that distribute power to Load rather than transfer bulk power across the interconnected system. LN's emanate from multiple points of connection at 100 kV or higher to improve the level of service to retail customers and not to accommodate bulk power transfer across the interconnected system. The LN is characterized by all of the following: a) Limits on connected generation: The LN and its underlying Elements do not include generation resources identified in Inclusions I2, I3, or I4 and do not have an aggregate capacity of non-retail generation greater than 75 MVA (gross nameplate rating); b) Real Power flows only into the LN and the LN does not transfer energy originating outside the LN for delivery through the LN; and c) Not part of a Flowgate or transfer path: The LN does not contain any part of a permanent Flowgate in the Eastern Interconnection, a major transfer path within the Western Interconnection, or a comparable monitored Facility in the ERCOT or Quebec Interconnections, and is not a monitored Facility included in an Interconnection Reliability Operating Limit (IROL). • E4 – Reactive Power devices installed for the sole benefit of a retail customer(s). Note - Elements may be included or excluded on a case-by-case basis through b) the Rules of Procedure exception process.
CIP Exceptional Circumstance	A situation that involves or threatens to involve one or more of the following, or similar, conditions that impact safety or BES reliability: a risk of injury or death; a natural disaster; civil unrest; an imminent or existing hardware, software, or equipment failure; a Cyber Security Incident requiring emergency assistance; a response by emergency services; the enactment of a mutual assistance agreement; or an impediment of large scale workforce availability.
CIP Senior Manager	A single official with the authority to bind the registered entity to adhere to the requirements within the CIP Standards, CIP-002 through CIP-015.
Control Center	One or more facilities hosting operating personnel that monitor and control the Bulk Electric System (BES) in real-time to perform the reliability tasks, including their associated data centers, of: 1) a Reliability Coordinator, 2) a Balancing Authority, 3) a Transmission Operator for transmission Facilities at two or more locations, or 4) a Generator Operator for generation Facilities at two or more locations.
Cranking Path	A portion of the electric system that can be isolated and then energized to deliver electric power from a generation source to enable the startup of one or more other generating units.

Cyber Assets	Programmable electronic devices, including the hardware, software, and data in those devices.
Distribution Provider	Provides and operates the “wires” between the transmission system and the end-use customer. For those end-use customers who are served at transmission voltages, the Transmission Owner also serves as the Distribution Provider. Thus, the Distribution Provider is not defined by a specific voltage, but rather as performing the distribution function at any voltage
Electronic Access Control or Monitoring Systems	Cyber Assets that perform electronic access control or electronic access monitoring of the Electronic Security Perimeter(s) or BES Cyber Systems. This includes Intermediate Systems.
Electronic Access Point	A Cyber Asset interface on an Electronic Security Perimeter that allows routable communication between Cyber Assets outside an Electronic Security Perimeter and Cyber Assets inside an Electronic Security Perimeter.
Electronic Security Perimeter	The logical border surrounding a network to which BES Cyber Systems are connected using a routable protocol.
Electronic Access Control or Monitoring Systems	Cyber System(s) that perform electronic access control or electronic access monitoring of the Electronic Security Perimeter(s) (ESP) or BES Cyber Systems (BCS), including those not protected by an ESP used by the responsible entity to convert routable protocol communications to non-routable communications to a BCS.
Element	Any electrical device with terminals that may be connected to other electrical devices such as a generator, transformer, circuit breaker, bus section, or transmission line. An Element may be comprised of one or more components.
External Routable Connectivity	Programmable electronic devices, excluding Shared Cyber Infrastructure, including the hardware, software, and data in those devices. Application containers are considered software of Virtual Cyber Assets (VCAs) or Cyber Assets. VCAs are not considered software or data of Cyber Assets.
Interactive Remote Access	User-initiated electronic access by a person using a bidirectional routable protocol: • To a Cyber System protected by an Electronic Security Perimeter(s) (ESP); • That is converted by the responsible entity to a non-routable protocol that allows access to a Cyber System; or • To a Management Interface. Interactive Remote Access does not include:

	• Communication that originates from a Cyber System protected by any of the Responsible Entity's ESPs; or • System-to-system process communication.
Physical Access Control Systems	Cyber Systems that control, alert, or log access to the Physical Security Perimeter(s) (PSP), exclusive of locally mounted hardware or devices at the PSP such as motion sensors, electronic lock control mechanisms, and badge readers.
Protected Cyber Assets	One or more Cyber Assets connected using a routable protocol within or on an Electronic Security Perimeter that is not part of the highest impact BES Cyber System within the same Electronic Security Perimeter. The impact rating of Protected Cyber Assets is equal to the highest rated BES Cyber System in the same ESP.
Shared Cyber Infrastructure	One or more programmable electronic devices, including the software that shares the devices' resources, that: • Hosts one or more Virtual Cyber Assets (VCA) included in a BES Cyber Systems (BCS) or their associated Electronic Access Control or Monitoring Systems (EACMS) or Physical Access Control Systems (PACS); and hosts one or more VCAs that are not included in, or associated with, BCS of the same impact categorization; or • Provides storage resources required for system functionality of one or more Cyber Assets or VCAs included in a BCS or their associated EACMS or PACS; and also for one or more Cyber Assets or VCAs that are not included in, or associated with, BCS of the same impact categorization. SCI does not include the supported VCAs or Cyber Assets with which it shares its resources.
System	A combination of generation, transmission, and distribution components.
Transient Cyber Asset	A Cyber Asset or Virtual Cyber Asset (VCA) that is: 1. Capable of transmitting or transferring executable code; 2. Not included in a BES Cyber System (BCS); 3. Not a Protected Cyber Asset (PCA) associated with high or medium impact BCS; and 4. Connected for 30 consecutive calendar days or less: • On a network protected by an Electronic Security Perimeter containing high or medium impact BCS; or • Directly (e.g., using Ethernet, serial, Universal Serial Bus, or wireless including near field or Bluetooth communication) to a: ▪ BES Cyber Asset; ▪ Shared Cyber Infrastructure; or

	▪ PCA associated with high or medium impact BCS. Virtual machines hosted on a physical Transient Cyber Asset (TCA) are treated as software on that physical TCA. Examples of TCAs include, but are not limited to, Cyber Assets or VCAs used for data transfer, vulnerability assessment, maintenance, or troubleshooting purposes.
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Commented [PH1]: RRC working glossary to be deleted prior to publication and consolidated into combined RRC Glossary document.